

indicator differs from its long-run mean. For example, if a bond currently appears 10 cents rich, and if its average degree of richness has been zero, and if the standard deviation of the rich/cheap indicator for that bond is 5.0, then a Z-score or T-stat³ is calculated for the bond of 2.0, by subtracting the mean from the current value and dividing the result by the standard deviation of the rich/cheap indicator. In this case, $\frac{10-0}{5} = 2$.

The clear problem with this approach is that it ignores the information the data contain regarding the speed with which the rich/cheap indicator tends to revert toward its mean. For example, if the rich/cheap indicator for the bond in our example bond appears to be independently and identically distributed each day, then our expected value for the rich/cheap indicator the next time this bond is observed is zero, regardless of the current value of the indicator.

On the other hand, if the rich/cheap indicator for our bond has exhibited slow mean reversion, say, with a half-life of 200 days, then our expectation for tomorrow's rich/cheap indicator for the bond should be virtually identical to today's value for the indicator.

Knowing the Z-score is 2.0 in this example tells us almost nothing about the value we should expect for the rich/cheap indicator for this bond tomorrow.

Mean Reversion

In most cases, there are strong *a priori* reasons for believing that rich/cheap indicators for bonds tend to be mean reverting. And in our experience, the mean reversion of these indicators is confirmed as an empirical matter as well.

As a result, we suggest applying the mean reversion models of Chapter 2 to the rich/cheap indicators obtained in these curve-fitting exercises.

For example, we find it useful to report the half-lives of these rich/cheap indicators, assuming they follow stochastic differential equations with linear drift coefficients, as with the Ornstein–Uhlenbeck process. More generally, we find it useful to calculate the ex ante Sharpe ratios of the rich/cheap indicators on various horizons, such as two weeks.

³The term *Z-score* typically refers to the case in which a demeaned normal random variable is standardized by a *known* standard deviation, in which case the resulting variable is also normal and can be compared to values in a standard normal table. The term *T-stat* refers to the case in which a demeaned normal random variable is standardized by a standard deviation that must be *estimated*. In this case, assuming the observations have been drawn from a normal distribution, the estimate of the variance will have a Chi-squared distribution. And the ratio of a normal variate to the square root of a Chi-square variate has a student-t distribution. In applications, we almost always need to estimate the standard deviation, in which case the student-t distribution is more appropriate.

Applications

Trading Bond Switches and Butterflies

One of the main reasons for a relative value analyst or trader to identify rich and cheap securities is to identify attractive relative value bond trades. For a hedge fund trader, this might mean establishing a short position in one bond against a long position in a nearby bond. For a portfolio manager at a pension fund, this might mean selling an existing long position in one bond and using the proceeds to purchase a nearby bond considered to offer greater value. We'll use the generic term *bond switch* for each of these trades. When a cheap (rich) bond is purchased (sold) against positions in two other bonds, one with a lesser maturity and one with a greater maturity, we use the generic term *bond butterfly*.⁴

The implicit assumption often made when conducting rich/cheap analysis is that bonds that are currently rich should be expected to cheapen and that bonds that are currently cheap should be expected to richen. However, this need not be the case, and the careful relative value analyst will try to identify the reasons that a bond is trading rich or cheap so as to form a reasonable expectation about the prospects for the bond in the future. For example, a bond that has begun to be squeezed in the repo market may have a spot price that appears expensive relative to other bonds, but it's quite possible that the squeeze will intensify before it weakens, to the detriment of a trader who has taken a short position. To protect against this and similar possibilities, we suggest taking care to understand changes in relative valuations rather than merely assuming that bonds will revert to fair value. It's not always possible to identify the reasons that a bond is rich or cheap, and to be honest it's not always necessary. But it's certainly prudent to make the effort.

Selecting Issues for Trade Expression

The rich/cheap analysis described here is also useful in identifying specific instruments for expressing views. For example, let's imagine that an analyst performed a principal component analysis of the yield curve, as per Chapter 3, and decided to implement a curve steepening trade between the two-year (2Y) and 10Y sectors. Further, let's imagine that he analyzed the term structure of swap spreads, as per Chapter 12, and concluded that bonds are likely to cheapen to swaps in the 10Y sector. As a result, he decides to express the 10Y

⁴The intention is that these switches and butterflies will involve bonds within the same sector of the curve. Otherwise, a switch between bonds with significantly different maturities would result in a position with significant curve exposure.

leg of the trade by shorting a bond in the 10Y sector. At this point, he could use the rich/cheap analysis in this chapter, perhaps combined with the mean reversion analysis in Chapter 2, to identify the specific bond in the 10Y sector to short.

Predicting New Issue Pricing

It's sometimes useful to predict the pricing of a new issue even before it's announced. For example, there have been times when newly issued German Schatze have been CTD into the Schatz futures contract. These typically would be issued according to a regular schedule, so the yet-to-be-announced bonds could affect the pricing of the futures contracts.

To take a view on the fair value of the futures contract even before the new issue is announced in this case, one could forecast the coupon and maturity date of the bond and forecast the extent to which the bond was likely to trade rich to the curve at the time it was issued. This would allow the analyst to assume a spot price and a coupon for the bond. From these, a conversion factor and a basis point value could be computed. By assuming a repo rate for the new issue, the analyst then could calculate the presumed converted forward price for the new issue, and the pricing of the futures contract could proceed along the lines discussed in Chapter 7.

Creating Generic Notional Benchmarks

There are times when a relative value analyst would like to track the price or yield of a notional benchmark (i.e. a bond that doesn't actually exist). For example, we may want to compare the constant-maturity yield of a 10Y bond trading at par against the 10Y swap rate. In that case, we'll need to create a synthetic or *notional* bond for this purpose.

Having a fitted curve of discount factors allows us to solve for the size of a coupon that will cause the price of our notional bond to be equal to par. This can be done numerically, but it also can be done algebraically via the discount function.

In particular, if Z_{10} is the price of a 10Y zero-coupon bond that pays one dollar at maturity, and if A_{10} is the value of an annuity that pays one dollar on each of the payment dates for the notional coupon bond in question, then the size of the coupon, C , that will cause our notional bond to have a fair or fitted price equal to par can be determined by the equation

$$100 = 100Z_{10} + CA_{10}$$

We can solve the equation for C to get

$$C = \frac{100(1 - Z_{10})}{A_{10}}$$

CONCLUSIONS

Our main innovation in this chapter is to place the rich/cheap analysis in the context of a regression, which is the tool of choice when forming expectations conditional on a particular set of information. These days, computing speeds tend to be sufficient for performing nonlinear regressions numerically in a real-time trading environment, allowing us to use a set of models that provide better fits to the data than models of the past, many of which relied on linear equations and/or simplifications. Having real-time fitted curves in the context of a regression allows us to use the resulting regression residuals as real-time rich/cheap indicators.

As swap spreads are used by many market participants as rich/cheap indicators, we'll assess their suitability as an alternative to fitted curves in Chapter 17. In summary, we'll find that only SOFR-asset swap spreads are (almost) free from the deficiencies of other types of swap spreads as rich/cheap measure. As a consequence, SOFR-asset swap spreads of basis swapped bonds can be used as an alternative global RV measure instead of or in addition to fitting curves through basis swapped bonds.

However, the switch to SOFR as the reference rate has introduced structural breaks in the time series of swaps, as highlighted in the Preface. Hence, even if fitted curve models may not be the only alternative for rich/cheap analysis in the presence of SOFR swaps, they are of particular relevance for generating constant maturity government bond time series as an alternative to swap data. By contrast, EURIBOR swaps provide time series with less structural breaks for historical analysis in EUR, but no good alternative for rich/cheap assessments.

In addition, regression residuals versus a fitted curve are preferable to swap spreads as rich/cheap indicators in a time-series context as well. For example, the means of our regression residuals are presumed to be zero. As a result, we can model the residual of each bond as a mean-reverting process over time, allowing us to characterize the speed with which bonds tend to return to their fitted values.

In contrast, the fair swap spread for a particular bond can change over time, for all the reasons cited in Chapter 17, which include unobservable

factors, such as the cost of equity capital for banks. As a result, it's difficult, if not impossible, to quantify the speed with which misvaluations correct when these misvaluations are identified using swap spreads. This is yet another reason we suggest using fitted curves rather than swap spreads as relative value indicators for bonds. Nevertheless, rich/cheap analysis can play a role in determining which swap spreads are likely to widen and which are more likely to narrow.

An Analytic Process for Government Bond Markets

INTRODUCTION

Having described several statistical and financial models separately, this chapter outlines a possible combination into an analytic process for government bond markets. At this stage, we'll limit the discussion to one single government bond market, for example, Bunds, and explain the additional elements from cross-currency relationships in Chapter 17.

In broad conceptual terms, the task of an analyst could be described as extracting the *relevant* information for trading and investing from the multitude of government bond prices. He could approach this task in two steps, i.e. on two levels of abstraction:

- Use fitted curves to generate constant maturity time series.
- Use PCA to extract factors from the constant maturity time series.

As explained in Chapter 3, PCA factors can guide maturity selection; once the maturities are selected, going back to the fitted curve model can help choosing the best specific bonds. For example, based on the PCA step-by-step guide in Chapter 3, the analyst may decide to enter into a long 5Y, short 2Y and 7Y butterfly on the Bund curve and then look at the rich/cheap figures from the fitted curve to select the cheapest 5Y and the richest 2Y and 7Y Bunds. In this set-up, the fitted curve is thus used both at the beginning of the analytic process, generating the input variables into the PCA, and at its end.

In addition, at various points of the analysis, mean reversion models can support the assessment of the relative risk and return of possible trades.

STEP 1: FITTED CURVES

The fitted curve model described in Chapter 8 provides several important functions in the analytical process:

- Its constant maturity time series are the input variables into (PCA) curve analysis.

- Already in isolation, it identifies opportunities for “micro” bond switches, for example, between different 5Y Bunds, such as a deliverable and a non-deliverable issue, or 5Y Bunds and 5Y OBLs, etc. In combination with a curve trade, this information can be used for asset selection at the end of the process.
- By using our approach to simultaneously calculate the fitted curve and adjust for external explanatory variables, such as issuance size, both the fitted curve and the rich/cheap measure are not affected by external influences. And these isolated external influences can be analyzed separately, for example, by looking at a time series of the benchmark premium. This also allows incorporating market information, for example, about flow and benchmarks, into the analytic framework.

STEP 2: PCA FOR MATURITY SELECTION AND CURVE TRADES

Using the constant maturity time series as input, PCA adds another level of abstraction and allows analysis of the Bund market via its uncorrelated factors. Rather than dealing with 100 specific bonds, we can assess the overall level, non-directional slope, etc. of the Bund market in general and base our curve trading on this aggregated information. For example, a long-only fund can look at 1-factor PCA residuals in order to identify the cheapest maturity segment, given the current overall yield level.

In addition, as the external variables of the fitted curve model allow the incorporation of market information (such as benchmark status), the factors allow the analysis to be linked to macroeconomic variables, as illustrated in Table 3.1. One could, for example, assess the impact of a change in inflation on the PCA factors and hence on the trading strategy.

The factors of a PCA are also a suitable input for automatically monitoring market opportunities. For instance, one OU process could be calibrated to each of the first three factors of a PCA on the Bund curve and return a signal if a pre-defined expected Sharpe ratio is exceeded. We have monitored the global markets for trading opportunities in this manner for many years, running PCAs on a variety of different input variables (commodities, currencies) and assessing their factors automatically via an OU process. Of course, this highlights only candidates for possible trades, which require further analysis – and maybe also other mean reversion processes than OU. However, due to its ease of implementation, OU remains a natural choice for monitoring a large number of possible trades in real time.

STEP 3: FITTED CURVES FOR BOND SELECTION

Working backwards from higher to lower levels of information aggregation, the fitted curve model can now again be used to select the cheapest or

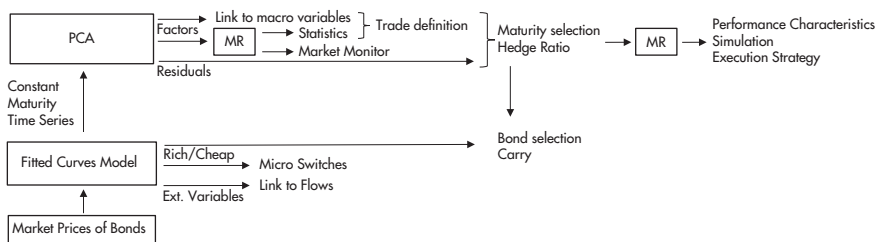


FIGURE 9.1 Combination of models into a possible analytic process for government bond markets.
Source: Authors.

richest bonds in the maturity segments bought or sold – adjusting for the effects of external influencing variables. This corresponds to step 11 of the guide in Chapter 3.

Once the individual bonds have been determined, the PCA results can be transferred to this level, corresponding to step 12 of the guide in Chapter 3:

- The PCA hedge ratios, which have been calculated on constant maturity time series, as outlined in Chapter 3, need to be adjusted to the BPVs of the individual bonds.
- With the hedge ratio of the individual bonds, the carry can be calculated.

By contrast, in order to avoid rolldown problems, the mean reversion characteristics should still be assessed via constant maturity time series. And at this stage of the analytic process, it is advisable to consult other models than OU as well: as mentioned in Chapter 2, the assumption of a constant variance may lead to overestimating the Sharpe ratio. Hence, assessing the mean reversion characteristics of the specific trade could be done via the bucketing method, for example, which brings elevated levels of variance at elevated levels of the trade to the attention of the analyst.

The (bucketing) mean reversion model calibrated to the trade can now be used to assess its expected performance characteristics, such as half-life and first passage time density, and to find the best execution strategy (target, stop loss, scaling), as outlined in Chapter 2. Moreover, the multi-variate mean reversion models discussed in Chapter 4 offer insights into the correlation characteristics, thereby providing also a framework for estimating the impact of adding the trade to an existing portfolio of positions.

Figure 9.1 attempts a graphical summary, with “MR” denoting the relevant mean reversion models.

As an alternative to the process outlined here, where MVOU is treated as a possible complement in step 3, it could replace PCA as the central element in step 2. In this case, the correlation and mean reversion characteristics of the constant maturity time series from step 1 are modeled directly via MVOU and the signals for potential trades are given by the deviations from its forecasts. And again, the richness and cheapness of individual bonds versus the fitted curve can then support the asset selection process in step 3. While this approach has the benefit of using one uniform model-framework for steps 2 and 3, sticking with PCA provides the intuitive insights from eigenvector interpretation and the link between factors and macroeconomic variables. In the end, each analyst will have to find that combination of models in an analytic process which best fits his goals and his way of thinking about markets.

Overview of the Following Chapters

Asset Swaps, Basis Swaps, Credit Default Swaps and Their Mutual Influences

Chapters 10–18 discuss asset, basis, and credit default swaps as well as their combinations and mutual influences. The complexity of the subject is reflected in a rather complex structure of the following chapters, which contain many interdependencies. We therefore start the ‘swap block’ of chapters with a map intended to help guide the reader through that maze.

The key to analyzing swap spreads of bonds is to consider bonds as a swap between the fixed rate given by the coupon and the floating rate given by the repo rate. From that perspective, buying a bond and simultaneously financing it via the repo market are equivalent to entering into a receive coupon (fixed) versus pay repo (floating) swap, with the maturity of the swap being determined by the maturity of the bond (likewise for the payment frequency of the coupon; we assume overnight repo financing unless stated differently). Chapter 11 contains an overview of the repo market.

This key opens the door to understand and price asset swap spreads of bonds: conceptually, buying a bond means receiving the coupon and paying the repo rate, while an asset swap exchanges the coupon rate against the reference rate of the swap.

- For the hypothetical case of the reference rate of the swap being the (overnight) repo rate of the individual bond, the cash flows of the asset swap match those of the bond financed in the repo market – and hence the asset swap spread should be zero from the perspective of funding rates (additional perspectives such as credit exposure will be considered below).
- For the more realistic case of asset swapping a US Treasury, the reference rate of the swap is SOFR. Hence, the swap spread of US Treasuries can be considered as a repo–SOFR basis swap over the life of the bond. In other words, buying a US Treasury financed in the repo market and asset